

LC 939: Minimum Area Rectangle

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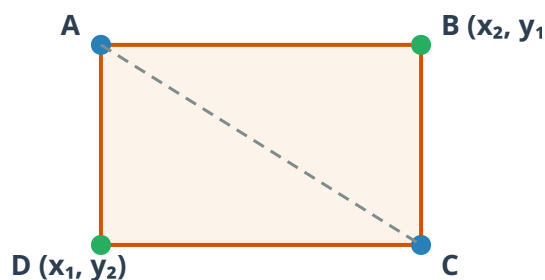
Target Companies: Flipkart, Goldman Sachs

Problem Statement

If no rectangle exist give 0, otherwise return minimum area, 2D coordinates points are given.

Logic & Intuition

- Now a rectangle has 4 corners.
- Naively we can choose 4 points in $O(n^4)$.
- But a rectangle is only defined by its diagonals.
- So let **aadya** be of A, **arpit** be of C (diagonals).
- Once A and C are decided, B and D are fixed.



Example Coordinates:

aadya = $(1, 1)$

arpit = $(3, 3)$

Means rectangle is:

$(1, 1) — (3, 1)$

$(1, 3) — (3, 3)$

- Hence one corner uses x from aadya, y from arpit $\rightarrow (1, 3)$
- Another corner uses x from arpit, y from aadya $\rightarrow (3, 1)$
- Suppose currently $(1, 1)$ and $(3, 3)$ then we need to see if $(1, 3)$ and $(3, 1)$ exists.
- So we can use set to see, as searching in set is $O(\log n)$ better than $O(n)$.

Code Implementation

```
class Solution {
public:
    int minAreaRect(vector<vector<int>>& points) {

        int n = points.size();
        set<pair<int, int>> st;

        for(auto p : points){
            st.insert({p[0], p[1]});
        }

        int ans = INT_MAX;

        for(int i = 0; i < n; i++){
            for(int j = i + 1; j < n; j++){

                int x1 = points[i][0];
                int y1 = points[i][1];

                int x2 = points[j][0];
                int y2 = points[j][1];

                if(x1 == x2 || y1 == y2){
                    continue; // lines are not rectangles
                }

                if(st.count({x2, y1}) && st.count({x1, y2})){
                    int w = abs(x2 - x1);
                    int h = abs(y2 - y1);

                    ans = min(ans, w * h);
                }
            }
        }

        if(ans == INT_MAX) return 0;
        return ans;
    }
};
```